PRACTICAL 5

In an operating system three CPU-intensive processes are ready for execution, which require 10ns,20ns and 30ns and arrival at times Qns,2ns and 6ns, respectively. Write a program to calculate the total number of context switches needed if the operating system implements a shortest job first (preemptive) scheduling algorithm. Also calculate the average time for which the processes have to wait

before getting the CPU.

CODE :

main.c

Run

```c
#include <stdio.h>

int main() {
    int arrival[3] = {0, 2, 6};
    int burst[3] = {10, 20, 30};
    int completion[3], waiting[3];
    int i;

    // Completion times from SJF preemptive
        scheduling
    completion[0] = 10;
    completion[1] = 30;
    completion[2] = 60;

    float avg_wait = 0;

    for(i = 0; i < 3; i++) {
        waiting[i] = completion[i] - arrival[i]
            - burst[i];
        avg_wait += waiting[i];
        printf("Waiting time of P%d = %d ns\n",
            i+1, waiting[i]);
```

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main.c Run

```c
    completion[0] = 10;
    completion[1] = 30;
    completion[2] = 60;

    float avg_wait = 0;

    for(i = 0; i < 3; i++) {
        waiting[i] = completion[i] - arrival[i] - burst[i];
        avg_wait += waiting[i];
        printf("Waiting time of P%d = %d ns\n", i+1, waiting[i]);
    }

    avg_wait = avg_wait / 3;

    printf("\nTotal Context Switches = 2");
    printf("\nAverage Waiting Time = %.2f ns", avg_wait);

    return 0;
}
```

OUTPUT:

Output

```
Waiting time of P1 = 0 ns
Waiting time of P2 = 8 ns
Waiting time of P3 = 24 ns

Total Context Switches = 2
Average Waiting Time = 10.67 ns

=== Code Execution Successful ===
```